

Alkali activation of fly ash. Part III: Effect of curing conditions on reaction and its graphical description

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ABSTRACT

The micro- and nano-structural characteristics of the reaction products of fly ash alkali activation depend, among others, on the curing conditions used: temperature, time and relative humidity. The present study focuses primarily on relative humidity. When the material is cured in air-tight containers, the silicon content of the initial aluminium-rich material gradually increases. This end product is dense and compact. When samples are in direct contact with the atmosphere, however, early age carbonation takes place, resulting in water loss and persistence of a high-aluminium content. This resulting material is granular and develops lower mechanical strength than the paste cured under high RH conditions. The main nano- and micro-structural differences observed in the materials as a result of different curing conditions have been shown through a graphical description of the reactions.

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1. Introduction

The alkali activation of fly ash is a chemical process in which the ash is mixed with certain alkaline activators and cured at a mild temperature to generate compact solids [1]. Very broadly, the activation process can be viewed as a series of destruction–condensation reactions in which initially unstable structural units subsequently form coagulated structures able to condense into the hydrated products (alkaline aluminosilicate hydrate, N–A–S–H gel) [2].

A number of models have been proposed in the literature to explain the formation of alkaline inorganic polymers (AIP). These models are based on results obtained with a variety of characterization techniques [2–5] and the existing knowledge about both silicate polymerization [6] and zeolite synthesis [7,6]. In 2005 Fernández-Jiménez et al. [9,10] proposed an updated version of the reaction mechanisms governing the alkali activation of aluminosilicates. Further to this model, the contact between the solid particles (aluminosilicates) and the alkaline solution induces the dissolution of the vitreous or amorphous component of the former, releasing (probably monomeric) aluminates and silicates. These monomers inter-react to form dimers, which in turn react with other monomers to form trimers, tetramers and so on. When the solution reaches saturation a N–A–S–H gel precipitates. Initially, this gel is an aluminium-rich aluminosilicate gel (Gel 1) [11]. Gel 1 formation may be explained by the higher Al³⁺ ion content in

the alkaline medium in the early stages of the process (from the first few minutes to the first 4–5 h). This, in turn, can be attributed to the fact that Al–O bonds are weaker and therefore more readily severed than Si–O bonds, making reactive aluminium more readily soluble than silicon [7]. As the reaction progresses, more Si–O groups in the solid particles dissolve, raising the silicate concentration in the medium. This gel gradually becomes richer in silicon, giving rise to a N–A–S–H gel, called Gel 2 [11].

The most prominent of the many parameters playing a role in N–A–S–H gel synthesis include the nature of the starting materials, the nature and concentration of the alkali activator and the curing conditions.

In prior research, Palomo et al. [4] observed that mechanical strength rose exponentially with temperature, substantially reducing the time needed to reach a certain value. They also reported the existence of time and temperature threshold values, however, at which mechanical strength rises more slowly, or may even decline. Temperature has also been observed to significantly affect the structural transition from the amorphous to the crystalline state in synthesized mineral polymers. Using XRD, Duxson [12] showed that even moderate rises in synthesis temperature led to visible increases in crystallinity in a polymer derived from metakaolin. Similarly, other authors have observed that in alkaline media the hydrothermal reaction may lead to the formation of semi-crystalline or polycrystalline phases [13–15]. The content of these (generally zeolite) phases grows with activation time [3].

In addition to curing time and temperature, relative humidity has been shown to play a role in the initial curing of alkali-activated fly ash cement. A study by Criado et al. [16] stressed the

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importance of using suitable relative humidity, for the rapid carbonation that may otherwise take place slows the rate of ash activation and the development of mechanical strength in the end product. Kovalchuk et al. [17] showed that curing conditions determine the amount of water available during the hardening of alkali-activated fly ash matrices.

In light of the technological relevance of curing conditions, then, further study of this aspect, in particular the effect of ambient relative humidity on alkaline activation, is fully justified. In addressing this question, the present study aims primarily to determine how the composition of the product of fly ash alkali activation and its nanostructure are affected by relative humidity and how these changes impact its physical and mechanical properties.

2. Experimental

2.1. Materials

The chemical composition of the Spanish fly ash used in the present study, acquired from the Compostilla power plant (Class F, ASTM C 618-03), is given in Table 1. A much more exhaustive characterization of the ash was reported by the authors in a prior paper [18].

The ash was activated with an 8-M NaOH solution, prepared from ACS-ISO 98% NaOH pellets supplied by Panreac S.A.

2.2. Method

The fly ash was mixed with the activating solution (liquid/solid ratio = 0.4) to prepare the pastes. The pastes were cured in an oven at 85 °C for 8 h, 20 h, 28 and 60 days. Some of the activated ash specimens (1 × 1 × 6-cm prisms) were tested for standard cement strength (as per European code EN 196-1) at those ages. The curing procedures summarized below were the same as used in Part I of this study [16].

- Method 1: the moulds containing the paste were placed in air-tight containers with a small amount of water (relative humidity > 90%). The containers were then stored in an oven at 85 °C.
- Method 2: the moulds were placed directly in the oven at 85 °C alongside a porcelain capsule containing water (relative humidity ≈ 40–50%).

After thermal treatment, the degree of reaction was determined by attacking the materials with 1:20 HCl (by volume) [19]. The HCl used to prepare the solution 1:20 was hydrochloric acid 37% PA-ACS-ISO. This procedure determines the percentage of reaction products (N–A–S–H gel and zeolites) generated at the different thermal curing times, which dissolve in the acid solution, and the percentage of unreacted ash, which remains in the insoluble residue. The dissolved fraction is a parameter that provides information on the degree of reaction (%).

The experimental process followed consisted on placing 1 g of the ground (to a powder, particle size < 65 µm) hardened material (activated ash) in a 250-ml flask containing 1:20 HCl. The mix was stirred with a plastic rotor for 3 h, after which it was filtered (through filter paper with a pore size of 15–20 µm) and the insoluble residue was washed with deionised water to a neutral pH. The

filter paper containing the residue was placed in a previously weighed platinum crucible, dried on a heat plate and calcined at 1000 °C in a furnace for 1 h. The percentage of the dissolved phases was deduced from the weight loss. The degree of reaction or “α” and “IR” or the insoluble residue were determined with Eqs. (1) and (2).

$$IR (\%) = \frac{P_{\text{initial}} - P_{\text{final}}}{P_{\text{final}}} \times 100 \quad (1)$$

$$\alpha (\%) = 100 - IR (\%) \quad (2)$$

where P_{initial} is the initial weight of the sample; P_{final} is the final weight of the insoluble residue after the HCl attack and calcination at 1000 °C; IR is the unreacted percentage of fly ash (insoluble residue) and α is the percentage of reaction product.

This was followed by microstructural and nanostructural studies of the products with XRD, SEM/EDX and MAS NMR. A PHILIPS PW 1730 diffractometer with Cu Kα radiation was used in this study, along with a JEOL 5400 electronic microscope fitted with an OXFORD LINK EDX microanalysis system (a total of 20 acquisitions were obtained from each sample for elemental analysis). The NMR studies were conducted with a BRUKER AVANCE-400 facility.

3. Results

Fig. 1 shows the degree of reaction of the working materials (under the two thermal curing procedures) against thermal curing time. Degree of reaction rose steeply in the Method 1 materials, whereas it barely varied with reaction time in the Method 2 materials.

Fig. 2, in turn, shows variations in compressive strength under the two working systems. The pastes cured at higher relative humidity (Method 1, moulds in air-tight containers) exhibited higher strength, even doubling the values found for the pastes cured at lower relative humidity. Mechanical strength rose visibly with curing time in Method 1 specimens, while it remained essentially flat in the Method 2 pastes.

The diffractograms for all the alkali-activated materials studied, as well as for the initial fly ash, are reproduced in Fig. 3. The diffractogram for the initial ash changed perceptibly when the ash was activated: the halo attributed to the vitreous phase of the original ash shifted slightly to higher 2θ values ($2\theta = 25\text{--}40^\circ$). This change denotes the formation of an alkaline aluminosilicate hydrate (N–A–S–H gel), the primary reaction product [3,19,20]. The minority crystalline phases (quartz, mullite and magnetite) detected in the initial material remained apparently unaltered. Alkaline activation also generated new zeolite-type crystalline phases. The type of these zeolites depended on the curing conditions: while with Method 1 hydrated sodalite ($\text{Na}_4\text{Al}_3\text{Si}_3\text{O}_{12}\text{OH}$, JCPDS 11-0401) and chabazite-Na ($\text{NaAlSi}_2\text{O}_6 \cdot 3\text{H}_2\text{O}$, JCPDS 19-1178) crystallized, only hydrated sodalite was detected in the Method 2 specimens. Moreover, the latter procedure favoured the formation of alkaline bicarbonates.

Fig. 4 shows certain morphological and microstructural characteristics of Method 1 and 2 activated ash after 20-h and 60-day curing. The presence of sodium aluminosilicate hydrate, the gel that forms the cementitious matrix, can be observed alongside the spheres of unreacted ash in these micrographs.

Table 1
Chemical analysis of the original fly ash.

	L.I.	I.R.	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	CaO	MgO	SO ₃	K ₂ O	Na ₂ O	TiO ₂	Total
Wt%	3.59	0.32	53.09	24.80	8.01	2.44	1.94	0.23	3.78	0.73	1.07	100

L.I. = loss on ignition; I.R. = insoluble residue.

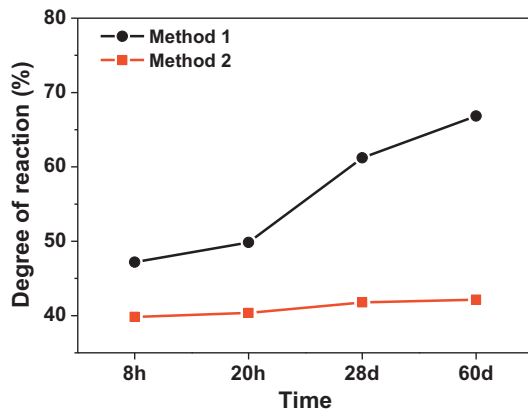


Fig. 1. Degree of reaction vs. curing time for two thermal curing systems.

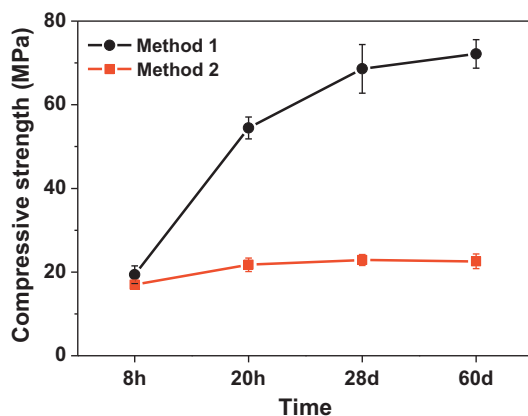


Fig. 2. Mechanical strength vs. curing time for two working systems.

After 20 h, the Method 1-cured ash had a generally porous matrix. The degree of reaction in this system was moderate, with many fly ash spheres still practically intact. Curing time proved to be an essential factor in the acquisition of mechanical consistency (Fig. 4(2)) and favoured the formation of small zeolite crystals (see Fig. 4(2a)). The main reaction product, N–A–S–H gel, became more silicon-rich over time, attaining an Si/Al ratio of 1.99 after 60 days (Table 2).

The 20-h micrograph of the Method 2 material showed a very porous, scanty compact matrix, with a great many unreacted ash particles apparently (more than in the Method 1 material). In

terms of reaction kinetics, this can be interpreted as follows. The matrices prepared pursuant to Method 2 developed more slowly than the Method 1 specimens. In the former, curing time did not appear to significantly affect the microstructure of the material, for the reaction barely progressed between 20 h and 60 days. Consequently, the matrix was less developed than in the Method 1 material. The presence of the alkaline bicarbonates detected in this system (Fig. 4(4b)) caused the material to develop a “granular” morphology in which the ash particles were partially attacked but conserved their initial spherical geometry. This same morphology has been observed in other alkaline cementitious systems when carbonates are added to the medium [21]. Moreover, no zeolites were observed in these matrices, which were scanty crystallized, as confirmed by XRD. Furthermore, the N–A–S–H gel forming in this system had a lower Si/Al ratio than the Method 1 material (see Table 2).

In the nanostructural study of the materials, the ^{29}Si spectrum of the original ash was found to be characterized by a very wide and scanty defined signal, indicative of the heterogeneous distribution of the silicon atoms (Fig. 5). The peaks identified at -80 , -84 , -95 , -100 , -104 and -109 ± 1 ppm were associated mostly with the vitreous phase forming part of the material [18].

Some of these signals (-80 , -95 and -100 ppm) were observed to disappear on the ^{29}Si spectra of the alkali-activated pastes cured for 8 h with the Method 1 procedure (Fig. 5(1)). Very intense peaks also appeared on these spectra at -88.2 , -93.2 and -98.5 ppm, along with lower intensity signals at -103.5 and -109.2 ppm, located in positions close to the ones detected for the silicon in certain zeolites [8,22]. These five signals were attributed to silicon atoms respectively surrounded by Al_4 , SiAl_3 , Si_2Al_2 , Si_3Al and Si_4 . The position of the signals did not vary significantly with increasing curing time (60 days), although peak resolution and intensity changed substantially. In the 8-h spectra, the most intense signals were observed at -88.2 , -93.2 and -98.5 ppm. The peak whose intensity grew most with curing was the one at -98.6 ppm. These differences in intensity can be associated with variations in the composition of the sodium aluminosilicate hydrate over time. At 8 h the gel formed, Gel 1, had a high-aluminium content [11], with a predominance of $\text{Q}^4(4\text{Al})$, $\text{Q}^4(3\text{Al})$ and $\text{Q}^4(2\text{Al})$ units. With time, the silicon content of this reaction product grew [11], whereby $\text{Q}^4(3\text{Al})$ and $\text{Q}^4(2\text{Al})$ units prevailed (signals at -93.3 and -98.6 ppm, respectively).

The 8-h ^{29}Si spectrum for the material cured at low relative humidity (Fig. 5(2)) contained an intense peak at -87.5 ppm, along with another series of signals at -100.3 , -105.9 and -111.0 ppm. These latter three signals, attributed to the vitreous phase in the starting material, indicated that the ash had barely reacted with the alkaline solution. The signal at -87.5 ppm, in turn, was associ-

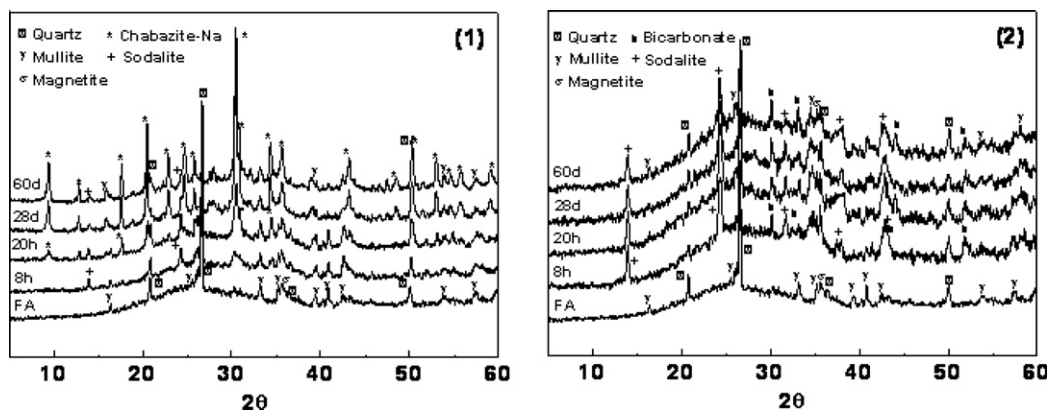


Fig. 3. Diffractograms of alkali-activated fly ash cured under Methods 1 and 2.

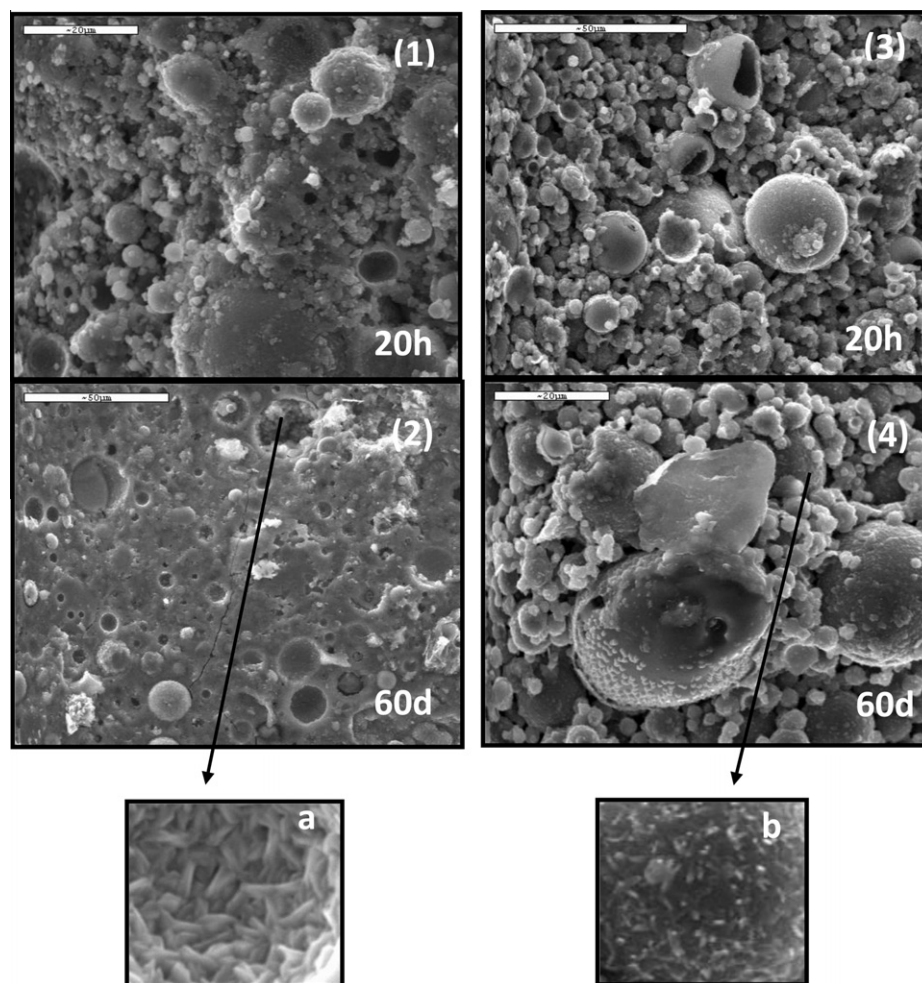


Fig. 4. Microstructure of the two systems studied: micrographs 1 and 2, cured under Method 1 for 20 h and 60 days, respectively; micrographs 3 and 4, cured under Method 2 for 20 h and 60 days, respectively.

Table 2
Mean Si/Al ratios in N-A-S-H gels deduced from microscopical analysis.

	Si/Al (20 h at 85 °C)	Si/Al (60d at 85 °C)
Method 1	1.88 ± 0.12	1.99 ± 0.08
Method 2	1.40 ± 0.09	1.68 ± 0.15

ated with the presence of $Q^4(4Al)$ units, apparently denoting the formation of an aluminium-rich reaction product (Gel 1). The area of this signal was greater than in the 8-h spectrum for the Method 1 material. The shape of the spectra barely changed with time (60 days): nonetheless, the intensity of the signals at -100.8 and -105.5 ppm declined slightly, which, together with the appearance of a signal at -93.5 ppm, provided evidence that the fly ash was gradually reacting. This spectrum exhibited much lower resolution than the 60-day spectrum for the high relative humidity material and bore a closer resemblance to the 8-h findings for Method 1. All of the foregoing confirms that unsuitable curing conditions may have a perceptible effect on fly ash activation reaction kinetics.

4. Discussion

Carbonation depends on the facility with which carbon dioxide can penetrate the pores of a material. Relative humidity plays a

fundamental role in carbon dioxide diffusion, for the diffusion coefficient of CO_2 in air is ten times higher than in water. Verbeck [23] showed that carbonation is favoured when relative humidity lies within the critical range of 60–70%.

Consequently, curing alkali-activated materials at a relative humidity of over 90% (Method 1) ensures pore saturation. According to previous works [24,25] and Kelvin–Laplace equation all pores are saturated with water except for pores above ≈ 10 nm in radius, justifying a weak CO_2 penetration, rendering carbonation practically nil. Under these conditions, the excess of available water and a high pH favoured dissolution of the vitreous component of the fly ash, stimulating the activation reaction and yielding a well developed material, mechanically speaking.

Water is a good reaction medium because solid aluminosilicate dissolution via alkaline hydrolysis consumes water. A certain amount of excess water is therefore needed to dissolve aluminate and silicate species. In a medium with high relative humidity the aluminate and silicate species are rapidly taken up in the aqueous phase and N–A–S–H gel begins to precipitate when saturation is reached. The water nominally consumed during dissolution is released in this condensation process. As a result, the time needed for a supersaturated aluminosilicate solution to form a continuous gel varies considerably with the curing conditions.

N–A–S–H gel must be regarded to be a zeolite precursor which over time gives rise to crystalline zeolites [3]. Zeolite stability is affected by the presence of chemical species such as water or organic

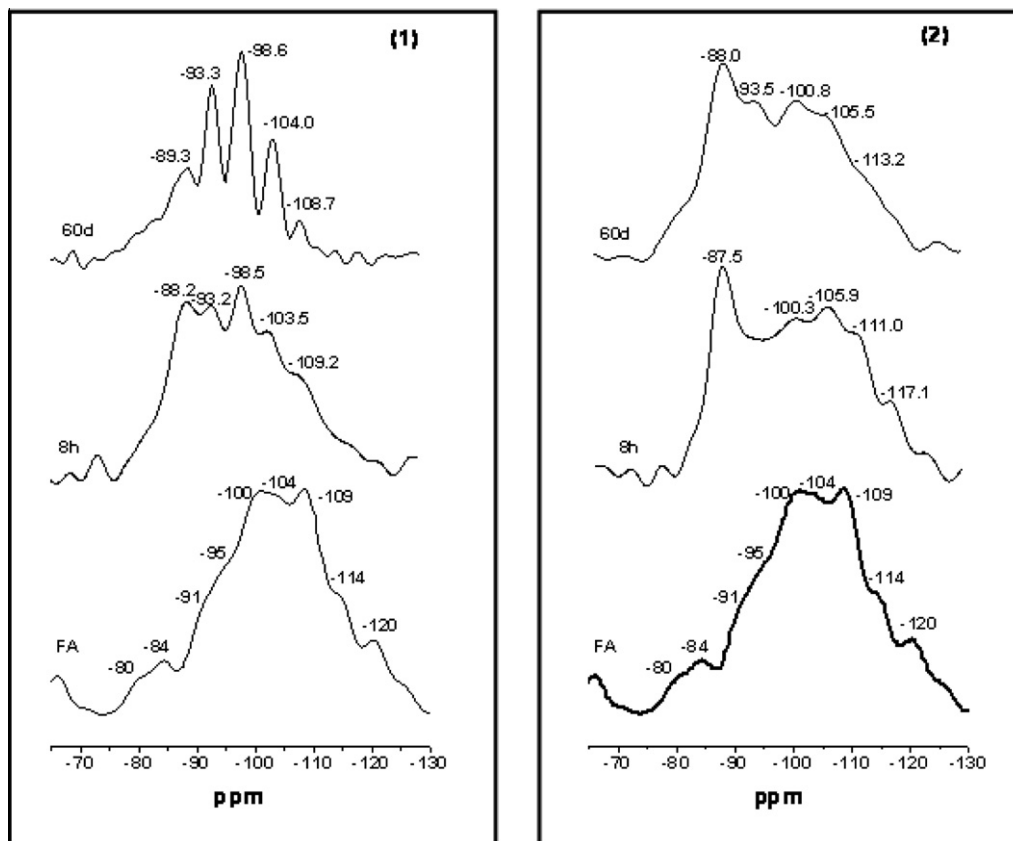
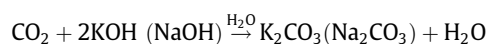


Fig. 5. ^{29}Si MAS NMR spectrum for alkali-activated Method 1 and Method 2 pastes.

molecules in zeolite cavities [26]. Regardless of their nature, these molecules lower the chemical potential of the zeolite and stabilize its structural grid. In the material cured here at a relative humidity of over 90%, the sodium cations that compensate for the charge deficit generated by the replacement of silicon with aluminium are hydrated and enhance zeolite stability. This in turn favours the crystallization of these minerals (hydrated sodalite and chabazite-Na) and increases their presence with time.

By contrast, when cured at a relative humidity of $\approx 50\%$, the materials are in a near critical situation in terms of carbonation. In such conditions, carbonation retards or halts the activation reaction, yielding a granular, porous and fairly weak product. The explanation for this change in reaction kinetics is that since the pores are not water-clogged, carbon dioxide molecules are able to penetrate and carbonate the material. Alkaline bases are known to carbonate rapidly when in contact with dissolved CO_2 .



In addition, curing in a medium with lower relative humidity where the amount of liquid water is somewhat scant retards the dissolution of the vitreous component of the fly ash (lower degree of reaction, see Fig. 1). These conditions would appear to have a greater effect on silicate than aluminate species, for aluminium dissolves more readily than silicon [7]. That may also explain the precipitation under such circumstances of an aluminium-rich aluminosilicate hydrate (N–A–S–H gel) whose composition (silicon content) barely varies over time because the decline in pH induced by carbonation hinders subsequent silicon dissolution (see Fig. 5(2)).

In such a low relative humidity system hydrated sodium cations are too scant to effectively favour zeolite stabilization. As a result, only hydrated sodalite forms (see Fig. 3).

The graphic-kinetic model proposed below on the grounds of the data compiled in this study charts the evolution of the phases in the starting materials and the products forming during alkali activation (Figs. 6 and 7).

As Fig. 6 shows, the starting material consists solely in fly ash, mullite and magnetite. Alkali activation induces the precipitation of a N–A–S–H gel, several zeolite species and alkaline bicarbonates (depending on the curing method). The same minority crystalline phases present in the original ash (quartz and smaller proportions of mullite) are detected throughout, along with a small number of fly ash spheres, some partially attacked.

After 8 h, the degree of reaction in the Method 1 material was 47.18% compared to 39.84% in the low relative humidity specimens. In other words, more N–A–S–H gel was generated in the former, where more ash reacted, than in the latter. The microstructure was similar in the two systems, with a porous and scanty compact matrix. Variation in matrix morphology with time depended on the curing method. When the material was cured under high relative humidity conditions, the end result was a dense, compact system and a strong matrix. By contrast, the microstructure of the system cured at a relative humidity of around 50% remained practically unchanged after 60 days in the oven at 85°C , when its matrix still contained a fairly large number of pores and a series of unreacted ash spheres (degree of reaction = 42.15%). Finally, in addition to N–A–S–H gel, zeolites were detected in both systems. The amount of hydrated sodalite and chabazite-Na, identified in the Method 1 system, grew with time. In the Method 2 system, alkaline bicarbonates and very small quantities of sodalite precipitated.

Fig. 7 illustrates the dissolution of the vitreous structure of the fly ash when it came into contact with the alkaline solution (8-M NaOH). The dissolved species subsequently inter-reacted to form more intensely polymerized structures that precipitated as N–A–S–H gel when they reached saturation. This figure relates the var-

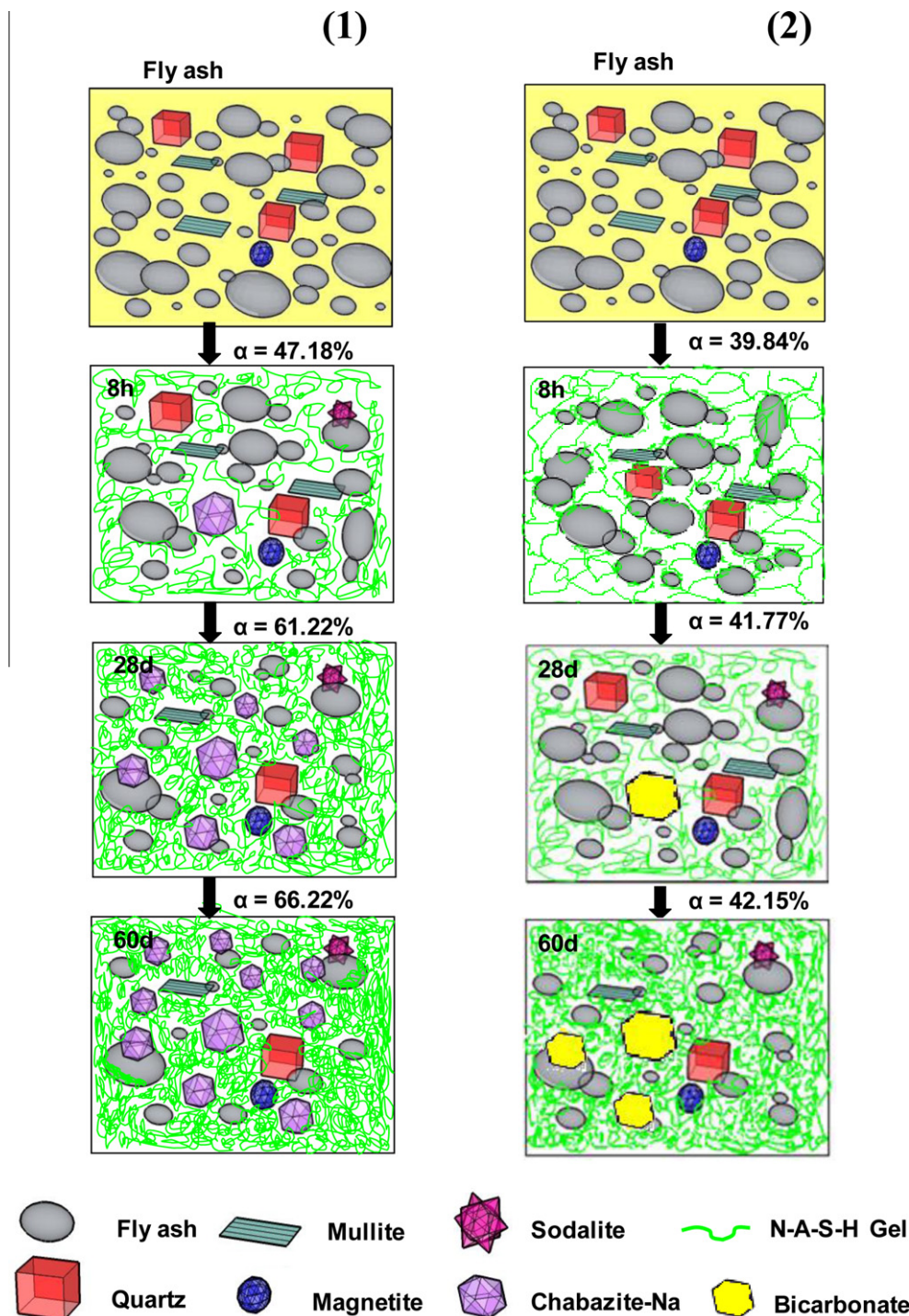


Fig. 6. Microscopic model of the working systems: (1) high relative humidity curing method; (2) low relative humidity curing method.

iation in N-A-S-H gel structure and composition with time to the curing conditions.

In the Method 1 material the dissolved species inter-reacted quickly, and after 8 h yielded an aluminium-rich gel (Gel 1) with a predominance of $Q^4(4Al)$ and $Q^4(3Al)$ units. The silicon content in this gel gradually rose with reaction time (thermal curing time). Since tetrahedral silicon tends to be surrounded by two or three aluminium atoms, the Si/Al ratio in the N-A-S-H gel grew (Si/Al ratio ≈ 2). This gel had a higher range structural order than the Method 2 material, as borne out by the higher resolution of the signals on the ^{29}Si spectra (see Fig. 5(1)). By contrast, in the material cured under the low humidity conditions that favoured initial car-

bonation, water loss in the material and a decline in pH, the vitreous phase of the fly ash dissolved more slowly when in contact with the alkaline solution. The aluminium-rich compound (Gel 1) initially formed, with a prevalence of $Q^4(4Al)$ and $Q^4(3Al)$ units, was less intensely polymerized than in Method 1. This phase barely changed over time due to the pH conditions induced by carbonation. Consequently, after 60 days, both its chemical composition and structure remained essentially unchanged (Gel 1: with an Si/Al ratio = 1.64, where the silicon tends to be surrounded by three aluminium atoms and exhibit low range structural order, as reflected in the poor resolution of the signals on the ^{29}Si spectrum).

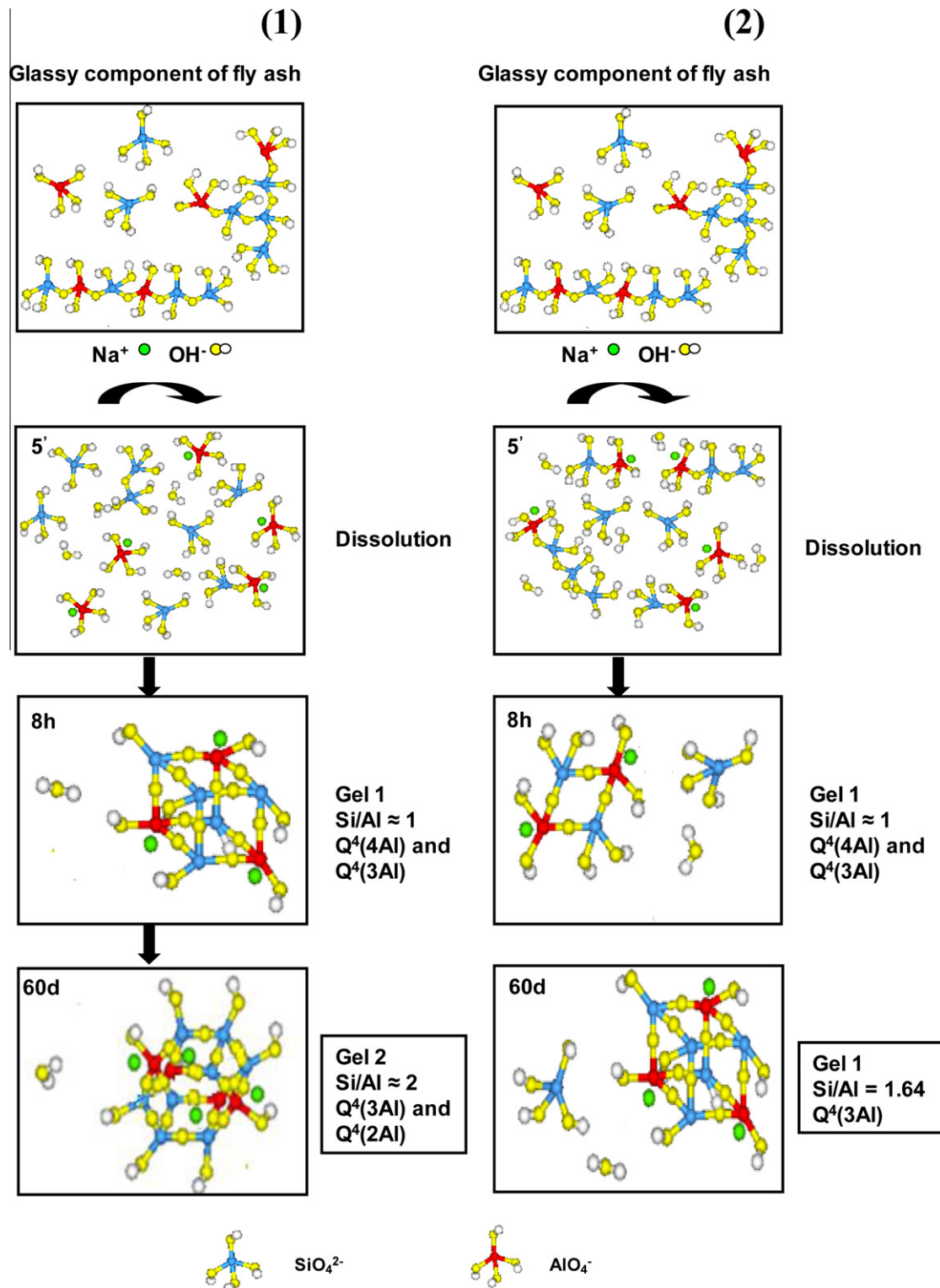


Fig. 7. Nanostructural model of the systems studied: (1) high relative humidity curing method; (2) low relative humidity curing method.

5. Conclusions

Curing conditions play a very important role in the micro- and nanostructural development of the reaction products obtained in fly ash alkaline activation.

Curing at a relative humidity of over 90%, in which the pastes are kept in air-tight containers, yields a dense, compact material,

whose initially high-aluminium content gives way to silicon up-take and good mechanical development over time.

Curing at low relative humidity with the pastes in direct contact with the atmosphere generates a granular, porous material. The aluminium-rich reaction products are very stable, with a chemical composition that remains unchanged throughout the curing process, ultimately resulting in a weaker material.

Finally, a graphical description of the main nano- and micro-structural differences observed in the materials as a result of employing thermal cured with different relative humidity conditions has been proposed.

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